

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

Lewei Xu

Supervised by Dr. Yihao Ding · Dr. Daochang Liu



Table of Contents

- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- Proof of Concept: A Simple Method
- Conclusion

Multi-Page Visually Rich Documents

Meaning is carried jointly by **text, vision, and layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.

[illegible]

Research paper	Signed form
born-digital	scanned, handwritten

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.

- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.

Financial report
dense tables

Product brochure
photo-led

- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

What is Document Understanding?

Recover **what is on each page** and **how the whole document is organised**, then answer questions over it.

LAYOUT ANALYSIS

3

PEW RESEARCH CENTER

Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

Yet many economic indicators show few gains for the community since the Great Recession

Latinos have become considerably more upbeat about their personal finances and optimistic about their financial future since the Great Recession, according to newly released results from a national survey of Latino adults. The survey also shows that Latinos have pulled even with the general U.S. population in their views of their personal finances and continue to outpace them on optimism about the future. However, community economic indicators show limited progress since the Great Recession.

Four-in-ten Latinos say their personal finances are in "excellent" or "good" shape, a 17 percentage point increase since 2008, when only 23% made a similarly positive assessment of their finances (the Great Recession began in December 2007).¹ By contrast, the share of all Americans who have a similarly positive view of their finances remained essentially flat during the same seven-year period (41% in 2008 vs. 43% in 2015).

The survey also shows that Hispanics are more upbeat in their financial expectations for the upcoming year than they were in 2008. About eight-in-ten Hispanic adults (80%) say they expect their family's financial situation to improve in the next year, up from 67% who said the same in 2008. By comparison, the U.S. public is not as upbeat –

Hispanics more confident in personal finances than in 2008, see better economic times ahead

% who say their personal financial situation is in excellent/good shape

% who expect their family's financial situation to improve a lot or "some" in the next year

Year	Hispanic Excellent/Good (%)	General Excellent/Good (%)	Hispanic Expect Improve (%)	General Expect Improve (%)
2004	31	23	70	67
2008	41	41	67	67
2015	58	43	80	64

Note: Shaded area marks the Great Recession, which ran from December 2007 to June 2009.

Source: National Survey of Latinos conducted Oct. 23-Nov. 30, 2015; surveys conducted Dec. 8-13, 2010; Oct. 24-28, 2012; Sept. 12-16, 2013; Dec. 7-11, 2011; Dec. 3-8, 2008; Oct. 19-22, 2006; Aug. 5-10, 2004.

¹Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

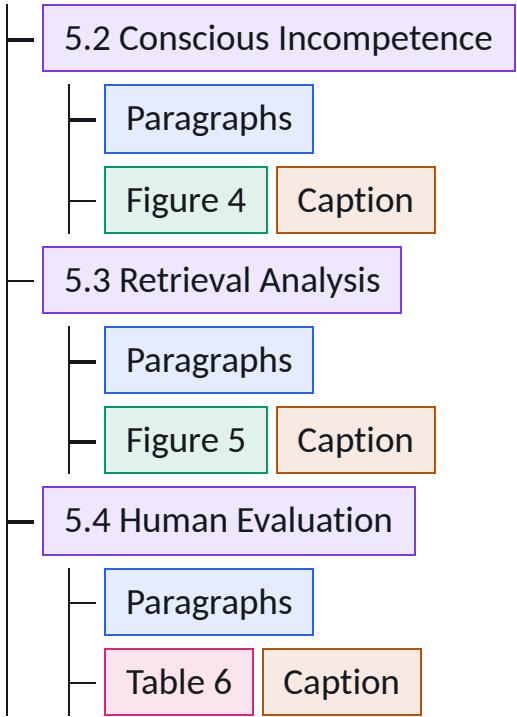
PEW RESEARCH CENTER

www.pewresearch.org

Title Section heading Text Figure / chart Table Caption / note Header / footer

DOCUMENT OUTLINE

Research paper, 24 pages



EXAMPLE QUESTION

“Does the model rated best by humans in Table 6 also have the best citation recall in Table 3?”

Evidence needed:

Table 6, this page

Table 3, previous page

Cross-page: neither table answers it alone.

LLM Assistants

Provide a demonstration of how you would use tool calls to answer a difficult question over this long visually rich document.

Designing a sample tool-based walkthrough for a cross-page question.

I'll demonstrate on a question that needs a figure and a table: "**How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?**" First I'll map printed page numbers to PDF pages, then read the relevant pages as images.

Read a file, updated tasks ▾

Added task **Locate pages for Figure 4.2 and Table 4.2** ✓

Request

Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task Read figure and table pages visually ▾

Request

Description	Read PDF pages 60-61 and extract the bar heights and token statistics
-------------	---

Added task **Verify and answer**

Request

Description	Cross-check figure values against surrounding text, then answer

Completed task: Locate pages for Figure 4.2 and Table 4.2

Started task Read figure and table pages visually

Read Honours Lewei Xu.pdf >

Verifying the data points plotted across the chart.

Completed task Read figure and table pages visually

Completed task **Verify and answer**

- A question over a **100-page thesis**: the assistant plans, locates the pages, then answers.
- It **renders the pages and reads them as images**.

Read Honours__Lewei_Xu.pdf ▾

[illegible]

Table 4.1 MMLongBench-Doc accuracy (%) for published systems using open 7-8B vision-language backbones. Reported values are taken directly from the cited papers; “vis” indicates results reproduced by another study rather than taken in the system’s original paper. OCR: parser or OCR text is provided as input. Trained: task-specific fine-tuning or RL within the pipeline. ^aA text LLM operates alongside the VLM. ^bRetriever only. ^cThe document outline includes visual-element descriptions generated by Qwen-VL-235B.

and layout questions decline. Multi-page accuracy also remains a clear weakness at 37.8%, below Doc-Trace's 41.1%, indicating that the higher overall accuracy does not eliminate the underlying difficulty of integrating evidence across pages.

Adaptive evidence acquisition. Figure 4.2 shows that the pipeline does not uniformly consume its full page budget. The largest group terminates after six pages, the earliest point permitted by the stopping floor $k = 6$, while progressively fewer questions continue deeper into the retrieval ranking and a smaller group reaches the $B = 25$ limit. Resopened trajectories account for part of this longer tail, with some questions requiring one or multiple CONTINUE decisions after an initial synthesis attempt. This differs substantially from the annotated evidence distribution, where most answerable questions contain only one or two gold pages. The gap reflects the practical setting in which the locations of these pages are unknown and additional contextual pages may also contribute useful evidence.

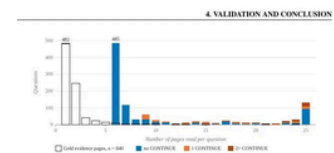


Figure 4.2 Trajectory of our method.

	Tokens			Latency (s)		Peak VRAM (GiB)	
	Turns	Input text	Input visual Output	Prefill	Decode		
reader	11.2	5,209	834	1.008	2.0	70.6	7.3
synthesizer	1.4	9,638	1,742	724	4.0	59.0	8.2
archiver	1.3	4,988	0	1.083	1.6	72.8	7.1
Per question, mean	13.9	78,215	11,733	13,721	29.7	968.1	8.2
Per question, median	9.0	37,484	7,272	6,081	13.8	544.6	8.0

Table 4.2 Inference cost of the pipeline per agent turn and question.

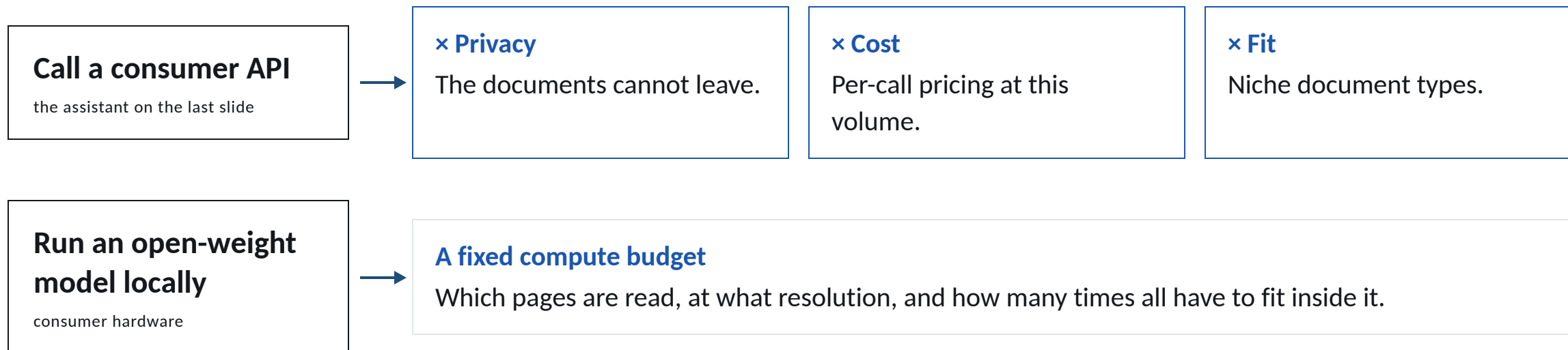
Inference cost. Table 4.2 quantifies the cost of this adaptive behaviour. The pipeline uses 13.9 turns per question on average, and accumulates a median of 9.6, with the reader accounting for 11.2 turns and the synthesiser and compiler invoked only 1.4 and 1.3 times per question respectively. Mean peak memory is 8.2 GB, resulting comfortably within the 16-GB budget of the deployment GPU, while the average question consumes 78.215 text input tokens, 11.733 visual tokens, and 13.721 output tokens across its trajectory. The principal cost is latency rather than memory: mean profile time is only 29.7 s per question per segment with 96.8 % of succeeding, with a median total of roughly 93 minutes. The method therefore trades additional sequential inference for bounded memory use and broader evidence acquisition, with the deployment parameters providing a direct mechanism for reducing trajectory depth where lower latency is required.

4.2 Open Challenges and Future Work

The survey of Chapter 2 identifies five challenges that define the open MP-VRDU frontier. They arise from a consistent pattern across the surveyed systems: progress comes from moving evidence handling out of the backbone, through compression, retrieval, or iterative orchestration, but each shift relocates the bottleneck rather than removing it. Chapter 3 and the system-level validation in §4.1 address parts of the first three challenges directly, but also show that their underlying trade-offs remain unresolved. Three concern system capability, in the evidence pipeline itself, in faithfulness, and in efficiency. Two concern data and evaluation, in supervision integrity and in evaluation scope.

Real-World Constraints

A hospital, a mining company, a law firm: **tens of thousands of documents** to classify, summarise, or extract from.



The research question: document understanding **under a fixed compute budget**.

Table of Contents

- Introduction

- **Survey of MP-VRDU**

Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding.

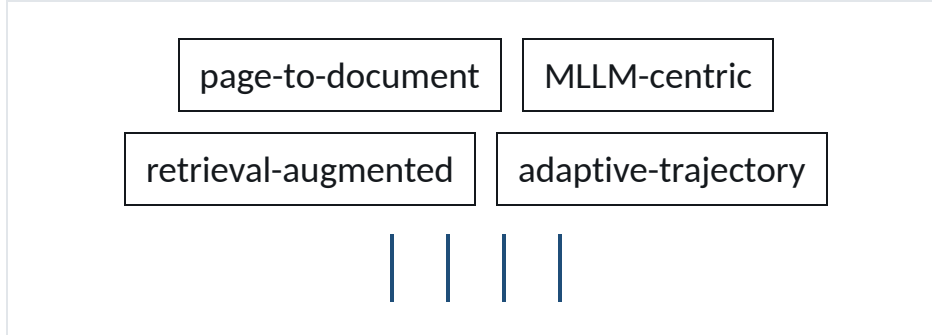
Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

[EMNLP 2026, main conference](#)

- Empirical Attribution Study
- Proof of Concept: A Simple Method
- Conclusion

Survey: Research Gap and Contributions

RESEARCH GAP



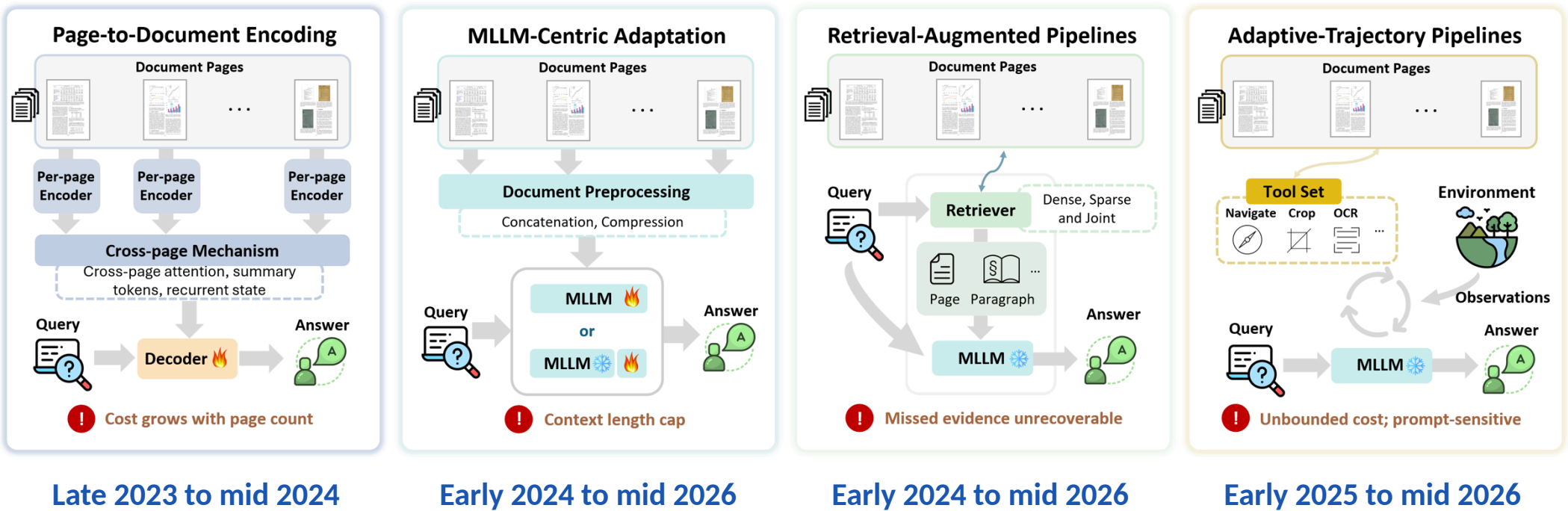
Multi-page systems grew along **parallel lines**, with inconsistent terminology. No survey treats multi-page as its own problem.

CONTRIBUTIONS

- 1 Defines multi-page understanding as a problem of **evidence management**.
- 2 A taxonomy: four architectural families, multimodal representation, inference-time and training strategies.
- 3 Consolidates the datasets and five open challenges.

Multi-Page Architectural Overview

The survey organises systems by **evidence management**, into four architectural families. **Page-to-document encoding models** jointly encode pages; **MLLM-centric adaptation models** fit multi-page input into an MLLM’s context and token budget; **retrieval-augmented pipelines** retrieve evidence before generation; **adaptive-trajectory pipelines** inspect evidence iteratively under an LLM controller.



Snowflake indicates typically frozen, fire indicates typically trained

Surveyed Systems at a Glance

Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Search	Reasoning	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VTS (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
MLLM-Centric Adaptation Models								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	-	CoT, HD	-
DocSeeker (2025)	CVPR	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B-Instruct	V	-	CoT, HD	-
Leopard (2024)	TMLR	No	SigLIP-SO400M	LLaMa-3.1-8B, Mistral-7B	V	-	CoT	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
URaG (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B / Qwen2.5-VL-7B	V	-	HD	-
Docopilot (2025b)	CVPR	No*	InternViT-300M	Multiple (Intern)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2.5-VL-7B-Instruct	V	-	-	-
DocSLM (2025)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	-
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025b)	CVPR	Yes	-	LLaMA3-8B, Qwen1.5-7B	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	-	CoT, PE, HD	-
Retrieval-Augmented Pipelines								
AVIR (2025)	MMAAsia	No	Pix2Struct	Qwen2.5-VL-3B-AWQ	V	D	-	-
DFVC (2026)	Electronics	No	VisRAG, ColQwen	Qwen2-VL-2B-Instruct	V	D	-	-
DREAM (2025)	MM	No	InternVL2 ViT	InternVL2-4B	V	J	HD	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Multiple (Qwen/Intern/Idefics)	V	D	-	-
MM-R5 (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	D	CoT	-
MoLoRAG (2025b)	EMNLP	No	ColPali	Multiple (Qwen/DS/LLaVA)	V	D, Nav	-	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	D	-	-
SLEUTH (2025)	preprint	No	ColPali, Qwen3-VL ViT	Multiple (Qwen/GLM/Gemini)	V	D	HD	4A
SV-RAG (2025)	ICLR	No	InternVL2, Phi-3-vision	Multiple (Intern/Pali/Phi)	V	D	-	-
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	D	-	-
CREAM (2024)	MM	Yes	Pix2Struct	LLaMA2-7B	T, V	J	HD	-
KGP (2024c)	AAAI	Yes	-	Multiple	T, L	S, IR, Nav	-	1A
MHier-RAG (2025)	preprint	Yes	Qwen-VL-Plus	Multiple (Qwen/DS/GPT)	T, V, L	J	CoT, HD	-
MLDocRAG (2026)	preprint	Yes	Qwen2.5-VL	Multiple (Qwen)	T, V, L	D	CoT, HD	-
MultiDocFusion (2025)	ACL	Yes	-	Multiple (Qwen/LLaMa/Mistral)	T, V, L	D	HD	-
M2RAG (2025a)	NeurIPS	Yes*	VisRAG-Ret	Multiple (Qwen)	T, V	J	-	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Multiple (Llama/Qwen)	T, V	J	-	5A
PDF-WuKong (2024)	preprint	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	J	-	-
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B-Instruct	T, V, L	D	-	-
VisDoMRAG (2025)	NAACL	Yes*	ColPali / ColQwen2	Multiple(Qwen/Gemini/GPT)	T, V	J	CoT	3A, SC
Adaptive-Trajectory Pipelines								
Doc-V* (2026)	preprint	No	Qwen2.5-VL	Qwen2.5-VL-7B-Instruct	V	D, IR, QR, Nav	ReAct, HD	1A
DocReact (2025a)	ACL	No	ColPali, VisRAG	GPT-4o	V	D, IR, QR	ReAct, QD	1A
MACT (2025)	preprint	No	Multiple	Multiple (Qwen/MiMo/Intern)	V	D	CoT, PE	4A, SC, SV
VRAG-RL (2026)	NeurIPS	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B-Instruct	V	D, IR, QR	ReAct, HD	1A
DocLens (2025a)	preprint	Yes	-	Multiple (GPT/Gemini/Claude)	T, V, L	J	CoT	4A, SA
DMAP (2026)	WWW	Yes*	ColPali	GPT-4o / Qwen-plus	T, V, L	J	HD	2A, SR
DocAgent (2025)	EMNLP	Yes*	-	Multiple (GPT/Gemini/Claude)	T, V, L	J, IR, Nav	ReAct, HD	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B-A3B	T, V, L	J, IR, QR	ReAct	1A
MM-Doc-R1 (2026)	preprint	Yes*	Qwen2.5-VL ViT	Qwen3-4B/8B	T, V	S, IR, QR, Nav	ReAct, HD	3A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Multiple (Qwen)	T, V	J, IR, QR	-	1A, SR, SV
ViDoRAG (2025a)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Multiple (Qwen/LLaMa/GPT)	T, V	J, IR	ReAct, HD	3A, SR, SV

1 Backbone

A **general-domain MLLM**, typically with **no extra training or fine-tuning**.

2 Modalities

Encoding models fuse **text, vision and layout**. MLLM-centric models are mostly **vision only**.

3 Search, fixed

Retrieval-augmented pipelines add one **dense or joint** retrieval step.

4 Search, adaptive

Adaptive-trajectory pipelines add **iterative retrieval, query reformulation and navigation**.

5 Reasoning

ReAct and **multi-agent** designs concentrate in the last family.

Page-to-Document Encoding: Hi-VT5

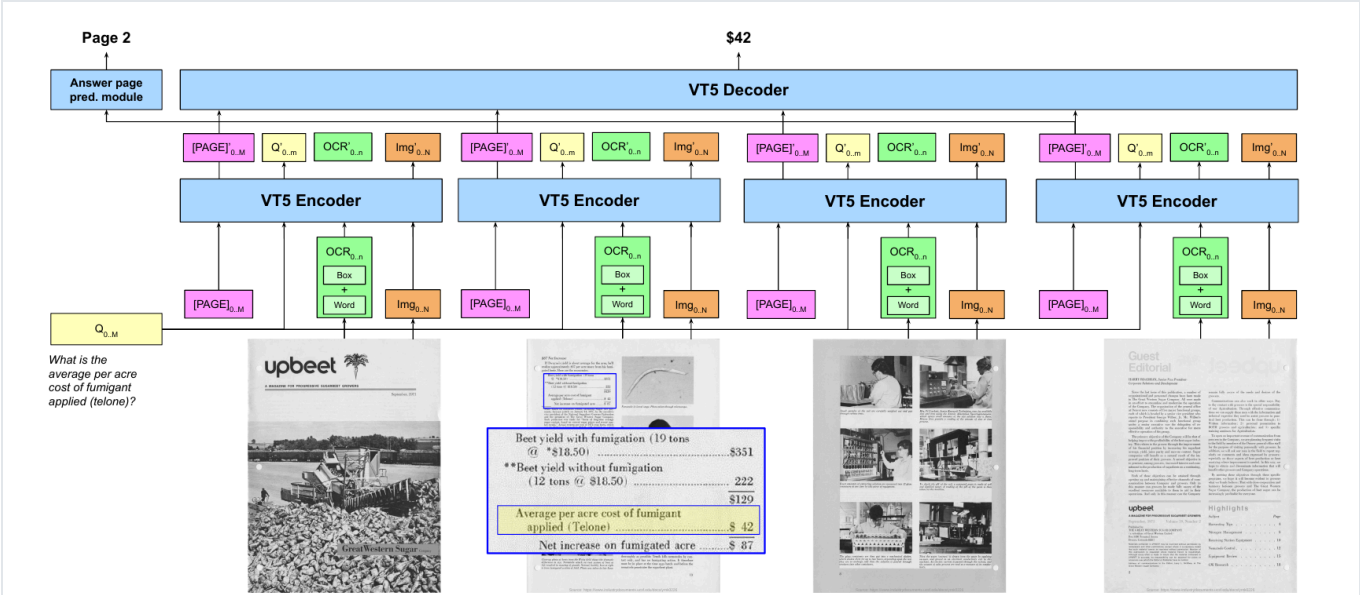
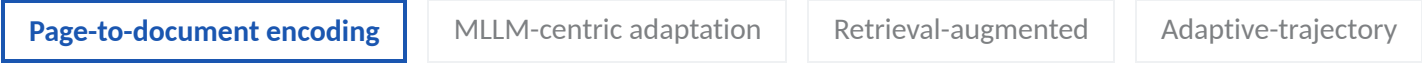


Figure from Tito et al., Pattern Recognition 2023

HOW IT WORKS

- Each page is encoded on its own from OCR text, layout boxes and image patches.
- Learnable [PAGE] tokens summarise each page. The decoder reads only those.

LIMIT

A page is squeezed into a few tokens, and **cost grows with page count**.

MLLM-Centric Adaptation: Docopilot

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

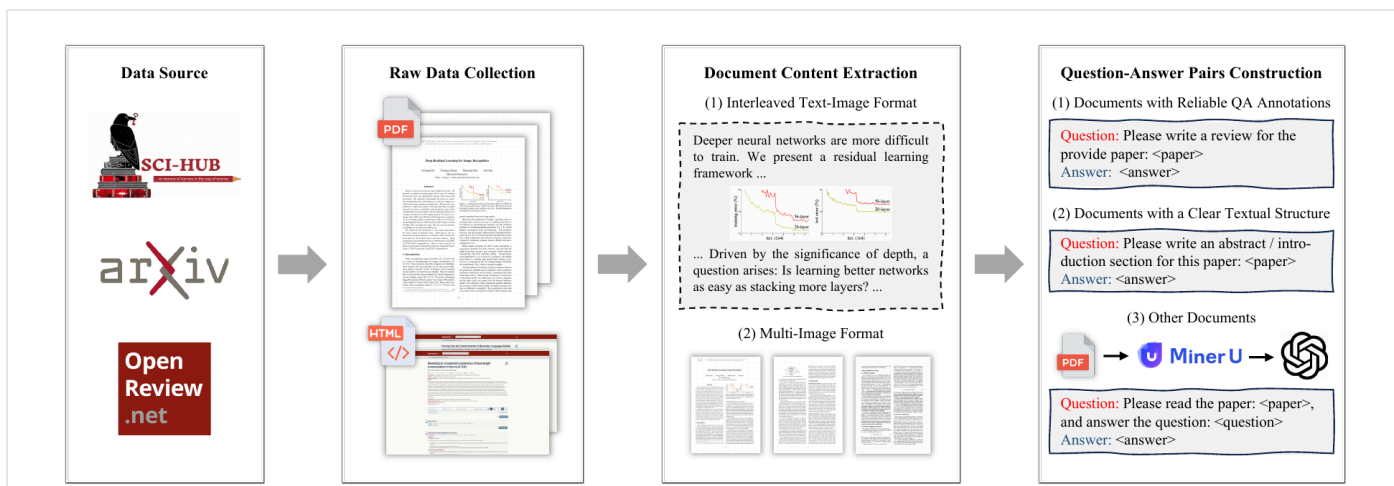


Figure from Duan et al., CVPR 2025

HOW IT WORKS

- A **native multi-page MLLM**: the whole document goes into the context, with no retrieval step.
- Trained on **Doc-750K**, built from papers on Sci-Hub, arXiv and OpenReview.

LIMIT

Bounded by the **context window**, and accuracy drops well before it is full.

Retrieval-Augmented: MoLoRAG

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

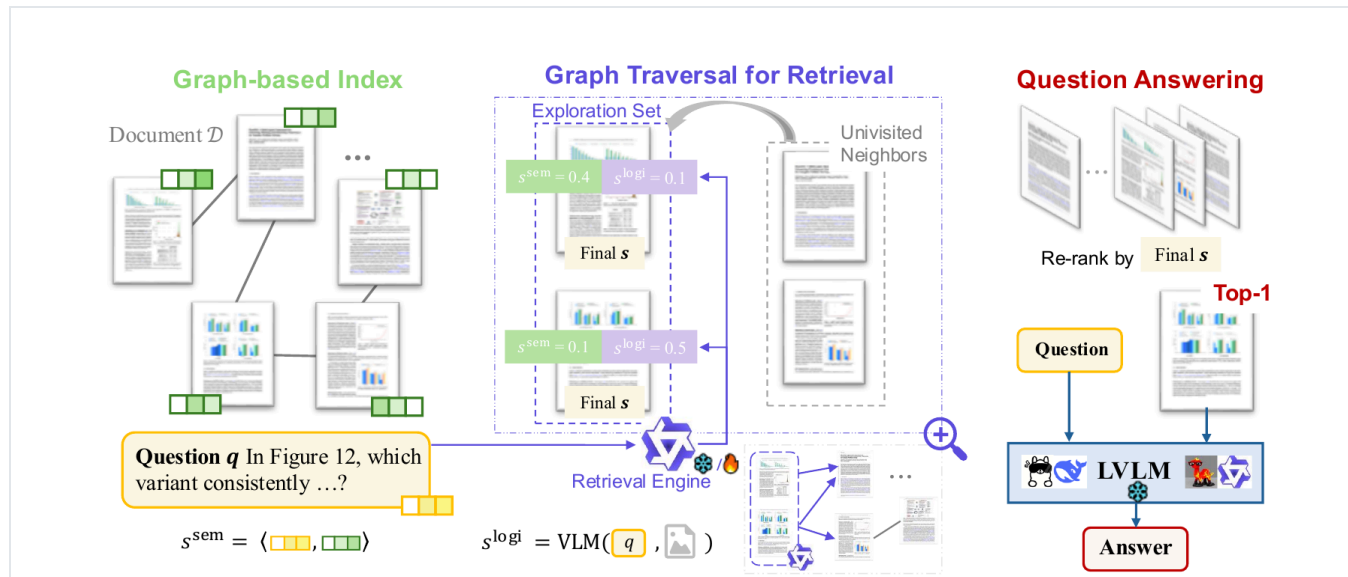


Figure from Wu et al., EMNLP 2025

HOW IT WORKS

- Pages form a **graph**. A VLM walks it from the most similar pages out to their neighbours.
- Each page gets a **semantic and a logical** relevance score. The top pages go to a frozen model.

LIMIT

Evidence the traversal misses is **unrecoverable** at generation.

Adaptive-Trajectory: DocLens

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

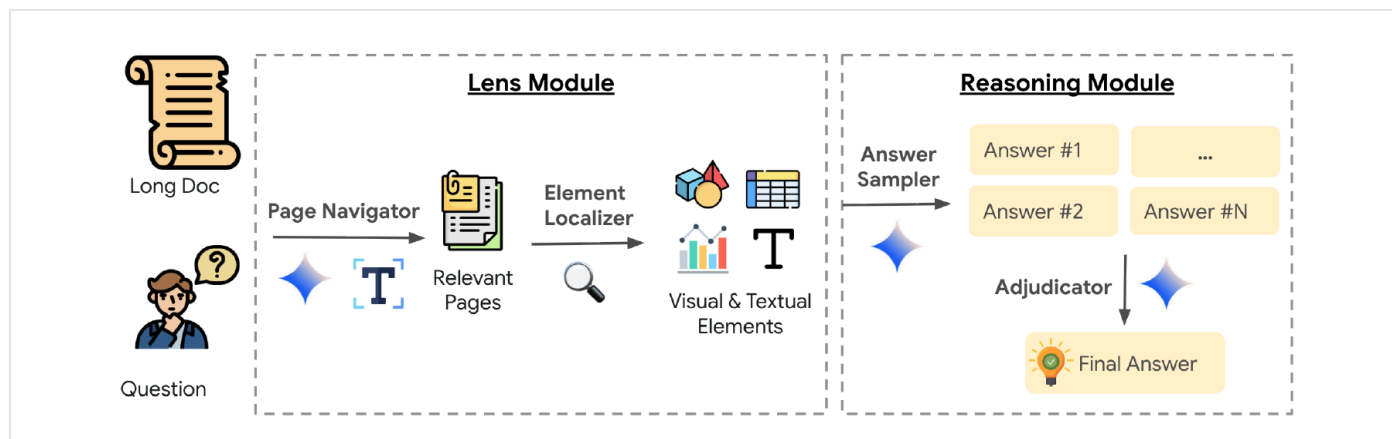


Figure from Zhu et al., ACL 2026

HOW IT WORKS

- **Lens module:** a page navigator finds the pages, then an element localiser zooms into figures, tables and text.
- **Reasoning module:** several answers are sampled and an adjudicator picks one.

LIMIT

Many model calls per question, so **cost is hard to bound**.

Table of Contents

- Introduction
- Survey of MP-VRDU

- **Empirical Attribution Study**

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

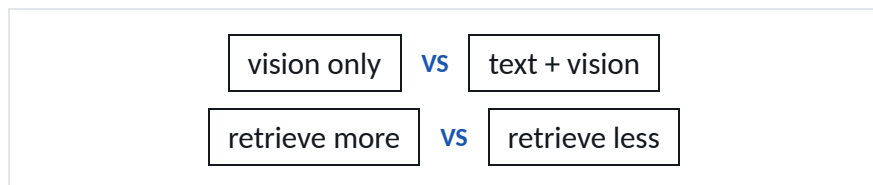
[Under review](#)

- Proof of Concept: A Simple Method
- Conclusion

Empirical Study: Research Gaps and Contributions

RESEARCH GAPS

- 1 **Competing design claims**, each tested on a different pipeline and dataset.



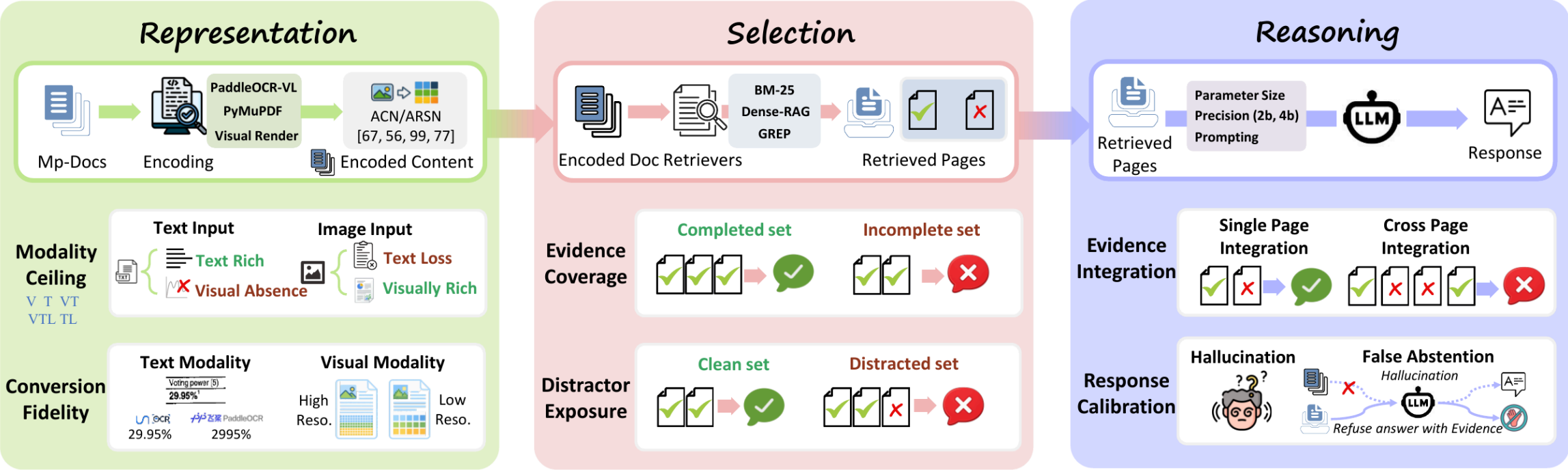
- 2 **No end-to-end empirical study** of where document understanding systems fail.

CONTRIBUTIONS

- 1 A **unified view of failure**: three architecture-independent loci, representation, selection and reasoning.
- 2 Controlled interventions in **one pipeline**, across document collections and independently developed model families.
- 3 **Practical guidance** for designing and deploying systems **under a compute budget**, from qualitative examples and accuracy-cost comparisons.

This is the local-deployment setting from the constraints slide.

Attribution Framework



REPRESENTATION

Was the information preserved when the page was converted?

SELECTION

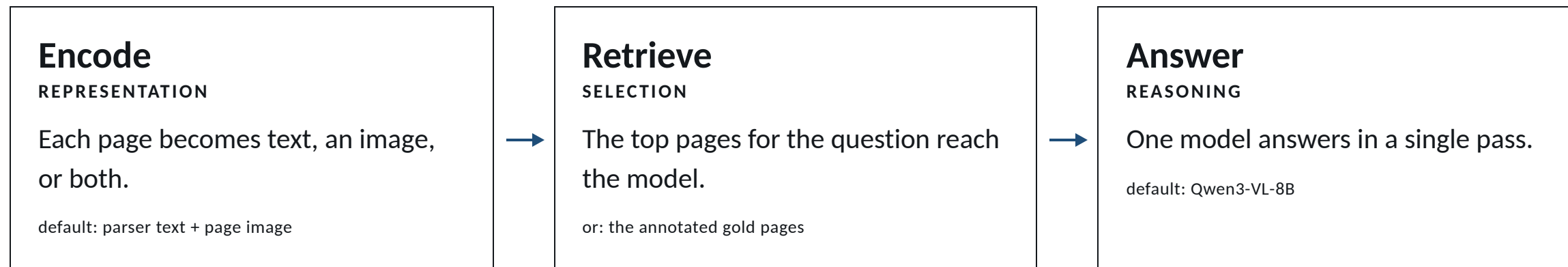
Did it actually reach the model?

REASONING

Did the model use it correctly?

Attribution by Construction

One simple pipeline. **Change one stage, hold the other two fixed.**



- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.

Representation Failures

Is the evidence in the input?

Modality ceiling

two inline icons, parsed as

Q What are the colors of the icons that the users touch to move the app up and remove the app respectively?

Gold gray, red

T ❌ "The document does not specify the colors"

TL ❌ "it does not mention any visual details"

--- ▼ **add the page image** -----

V ✓ "The icon is gray. ... The icon is red."

TLV ✓ "gray (a gray upward arrow) ... red"

Colour exists only in pixels; a text input leaves the reasoner nothing to read.

Conversion fidelity

scanned slide, text by parser

Q What is the name of the company that created this presentation?

Gold AVALAUNCH MEDIA

PaddleOCR ❌ "Asterks" (parse: KICK ASTERKS)

MinerU ❌ "does not provide any evidence"

--- ▼ **re-convert, or add the image** -----

Unlimited ✓ "AVALAUNCH MEDIA"

with image ✓ "Avalaunch Media ... in the banner"

The error sits in the transcript, not the reasoner, and each parser fails differently.

IMPLICATIONS

- **Text and vision are complementary.** Start from reliable embedded text, and add the page image when the question needs appearance or the parse looks wrong.
- Images are not free: about **26 s of prefill** per question with images against 1.8 s for text.

Selection Failures

Is the right evidence in context?

Evidence coverage	author list on pages 13, 14	Distractor exposure	10 gold pages of a 20-page deck
Q How many chapter authors of this report come from Columbia University? Gold 3 page 13 ✗ “two” (Horton, Kinney) page 14 ✗ “one” (Garcia-Pando) --- ▼ add the complementary page ----- both pages ✓ “three” (Horton, Kinney, Garcia-Pando) Each page is relevant, yet neither suffices: the answer is a property of the page set.		Q How many lateral bar graphs are present in the document? Gold 10 +3 pages ✗ “there are 12 lateral bar graphs” +5 pages ✗ “there are 13 lateral bar graphs” --- ▼ remove competing pages, or read on ----- gold pages ✓ “there are 10 lateral bar graphs” +7 to +18 ✓ “there are 10 lateral bar graphs” A few added pages inflate the count; the gold set, or more context, counts right.	

- IMPLICATIONS
- Select evidence as a set: judge whether the whole set is sufficient, not each page on its own.
- Retrieve broadly for recall, then filter or rerank what reaches the reasoner. Judge retrieval by end-to-end answers and latency, not recall alone.

Reasoning Failures

Is the evidence used correctly?

Evidence integration

a diagram, an ablation table

Q If I drop the green module shown in Figure 6, what is the absolute percentage drop on 100-shot TACREV setting?

Gold 0.5%

2B ❌ "2.4%" (reads another table)

4B ❌ "-9.6%" (right table, wrong row)

8B ❌ " $(75.9 - 74.9) / 75.9 \dots \approx 1.32\%$ "

--- ▼ a larger reasoner

32B ✅ " $75.9 - 75.4 = 0.5$ "

Each size breaks a different step of one chain: locate the table, read the cell, compute.

Response calibration

two tables, two pages

Q How much lower is the accuracy of GPT-4 on SituatedQA compared to MMLU Chemistry?

Gold 16.7%

abstain ❌ "Not answerable" (false refusal)

--- ▼ reason before deciding

abstain_cot ✅ " $79.9\% - 63.2\% = 16.7\%$ "

none ✅ "... 16.7 percentage points lower"

cot ✅ "Step 1: ... Answer: 16.7%"

The question is answerable; the abstain instruction turns multi-step difficulty into refusal.

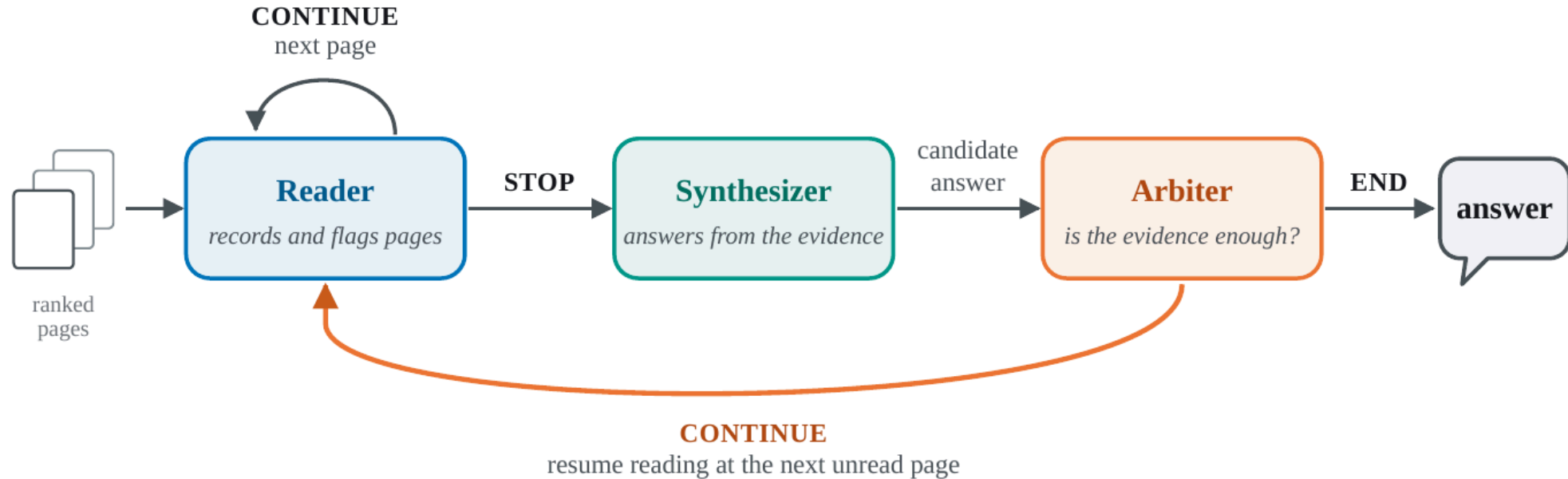
IMPLICATIONS

- **Match reasoner capacity to the operation** the task needs. Under a memory budget, a quantised larger model can beat a smaller one.
- **Tune refusal together with reasoning** on the deployed backbone, and report false refusals alongside accuracy.

Table of Contents

- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- **Proof of Concept: A Simple Method**
- Conclusion

From Findings to a Simple Method



REPRESENTATION

Every page is read as **text and image together**.

SELECTION

Read broadly, then pass on compact records, not full pages.

REASONING

A separate arbiter decides **whether the evidence is enough**.

Simple Method Results

Method	OCR	Trained	MMLongBench-Doc				LongDocURL			
			SIN	MUL	UNA	ACC	UND	REA	LOC	ACC
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7
Whole document (via DocTrace)	×	×	46.9	35.3	25.8	38.5	–	–	–	45.1
M3DocRAG (via SLEUTH)	×	×	–	–	48.6	45.9 ^a	60.1	52.4	45.5	54.6
DocAgent (via MARDoc)	✓	×	–	–	–	46.6	–	–	–	–
MDocAgent (via SLEUTH)	✓	×	–	–	49.7	47.8 ^a	57.5	51.5	44.7	53.1
MoLoRAG (via SLEUTH)	×	×	–	–	51.7	48.8 ^a	63.4	52.7	49.3	57.6
SLEUTH	×	×	–	–	67.4	52.8 ^a	65.7	53.0	53.6	60.0
DocTrace	✓	✓	53.2	41.1	70.5	52.9	–	–	–	56.4
Ours (4-bit)	✓	×	60.5	39.2	76.9	56.5	67.2	66.3	47.9	61.3

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

Conclusion

SURVEY

Organise

Multi-page document understanding is a problem of **evidence management**.

EMPIRICAL STUDY

Locate

Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

PROOF OF CONCEPT

Validate

The findings hold up in a **working, training-free system**.

Thank you Questions?

Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted around a frozen MLLM**, or **trained into it**.

Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Datasets and Evaluation

PRIMARY

MMLongBench-Doc

135	48.3	847	244
documents	mean pages	answerable	unanswerable

- Annotated with **evidence pages, evidence sources and answerability**, which the experiments need.
- 367 of the answerable questions need two or more pages.

COMPLEMENTARY

LongDocURL

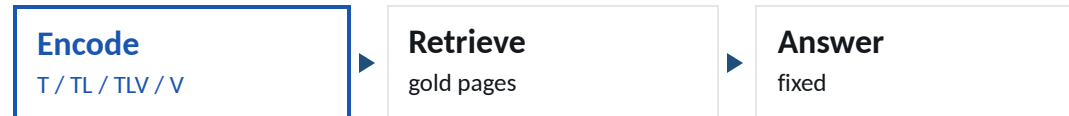
396	85.6	2,325
documents	mean pages	questions

- **Born-digital only**, so answers are often in the text layer, even for figure questions.
- Used for the modality-ceiling result only: its evidence pages are not a minimal set.

Judge: Gemini-2.5-Flash at temperature zero returns correct, incorrect or abstained. Agreement is **96.0%** with GPT-4o and **98.3%** with a human annotator.

Modality Ceiling

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
MMLongBench-Doc								
Text	23.0	11.5	7.3	8.0	36.9	44.4	51.0	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	49.2	43.2
Table	15.9	7.2	6.7	9.1	35.6	44.7	44.7	34.1
Chart	32.6	28.1	7.3	6.2	16.3	16.9	42.7	40.4
Figure	36.6	24.8	7.6	7.9	14.5	22.1	42.4	39.0
<i>All</i>	26.3	17.2	6.2	7.1	29.0	35.7	49.5	42.6
LongDocURL								
Text	4.6	3.5	1.5	1.6	71.8	75.6	78.2	70.0
Layout	2.1	1.6	0.7	0.9	45.0	49.2	51.3	49.3
Table	5.0	3.2	1.1	0.8	57.7	65.6	63.6	52.4
Figure	6.7	8.6	1.5	1.3	59.9	62.8	69.9	61.9
<i>All</i>	4.7	4.3	1.3	1.3	60.3	64.5	67.3	59.4



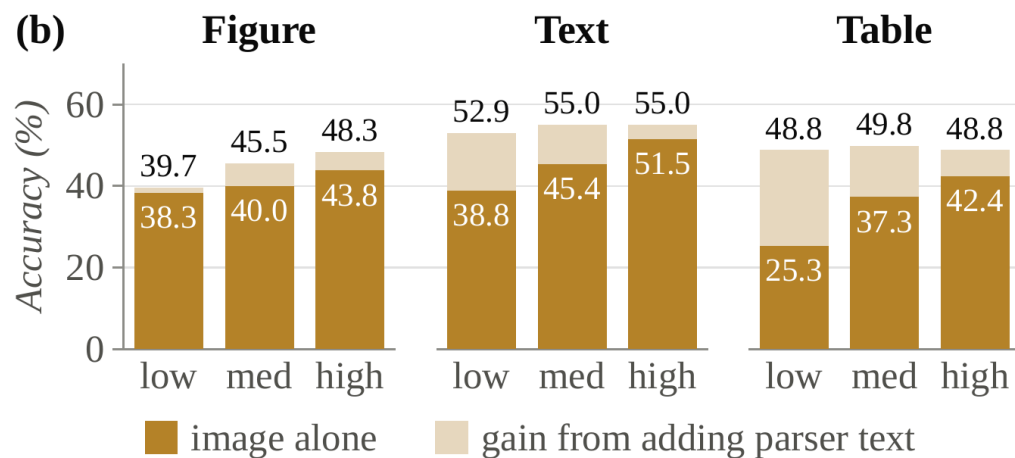
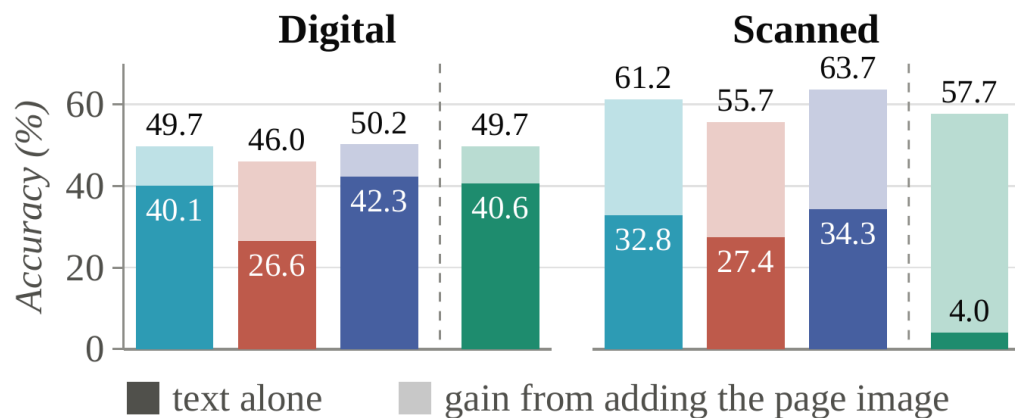
REPRESENTATION

Evidence accessibility depends on modality

- On MMLongBench-Doc, text-only input refuses **chart and figure** questions most. Page images bring refusal **under 10%**.
- Images alone stay below text plus images: 42.6% against 49.5%.
- On born-digital LongDocURL, adding vision to parser text gives only **2.8 points** overall, 7.1 on figure questions.

Conversion Fidelity

(a) PaddleOCR MinerU Unlimited-OCR PyMuPDF



Encode
parser, resolution

Retrieve
gold pages

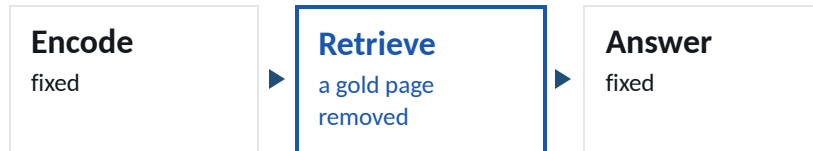
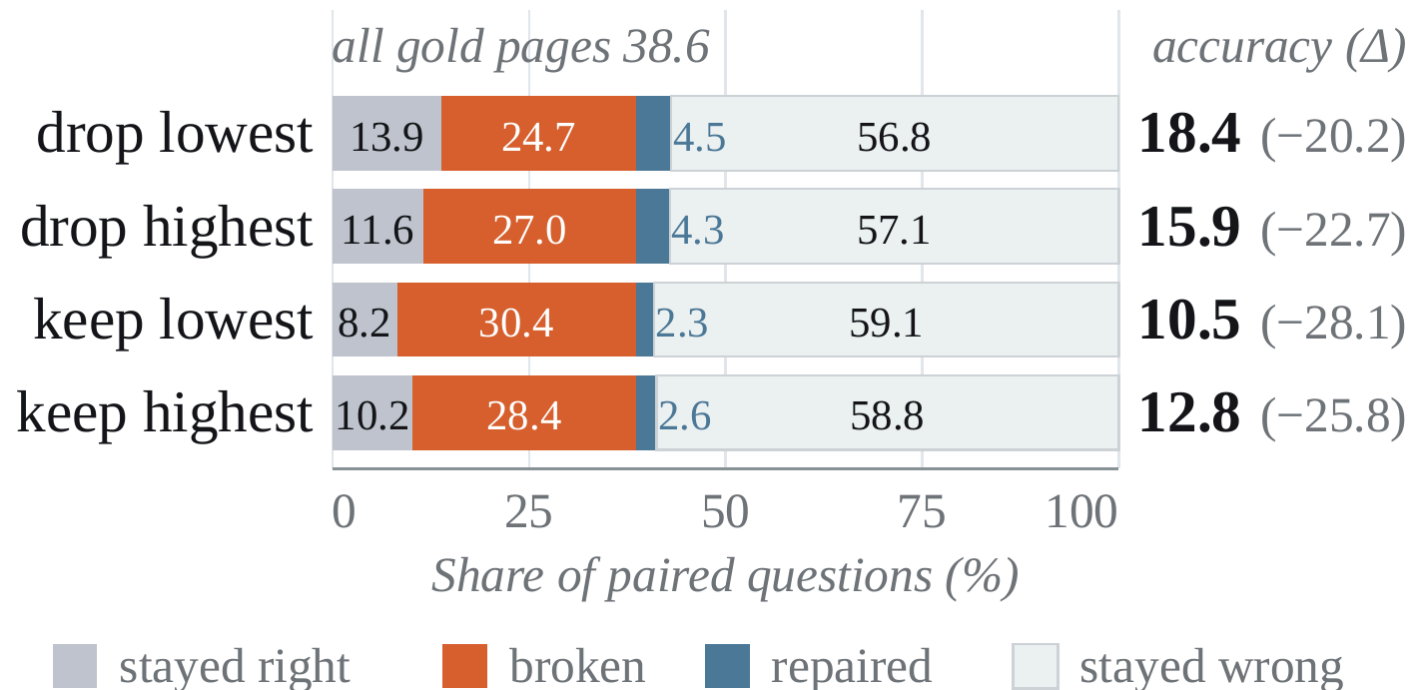
Answer
fixed

REPRESENTATION

Complementary modalities reduce conversion sensitivity

- Adding page images cuts the spread across parsers from **15.7 to 4.2 points** on digital documents.
- Higher resolution helps images alone far more than text plus images.
- Scanned documents stay sensitive to the parser, and figure questions to resolution.

Evidence Coverage

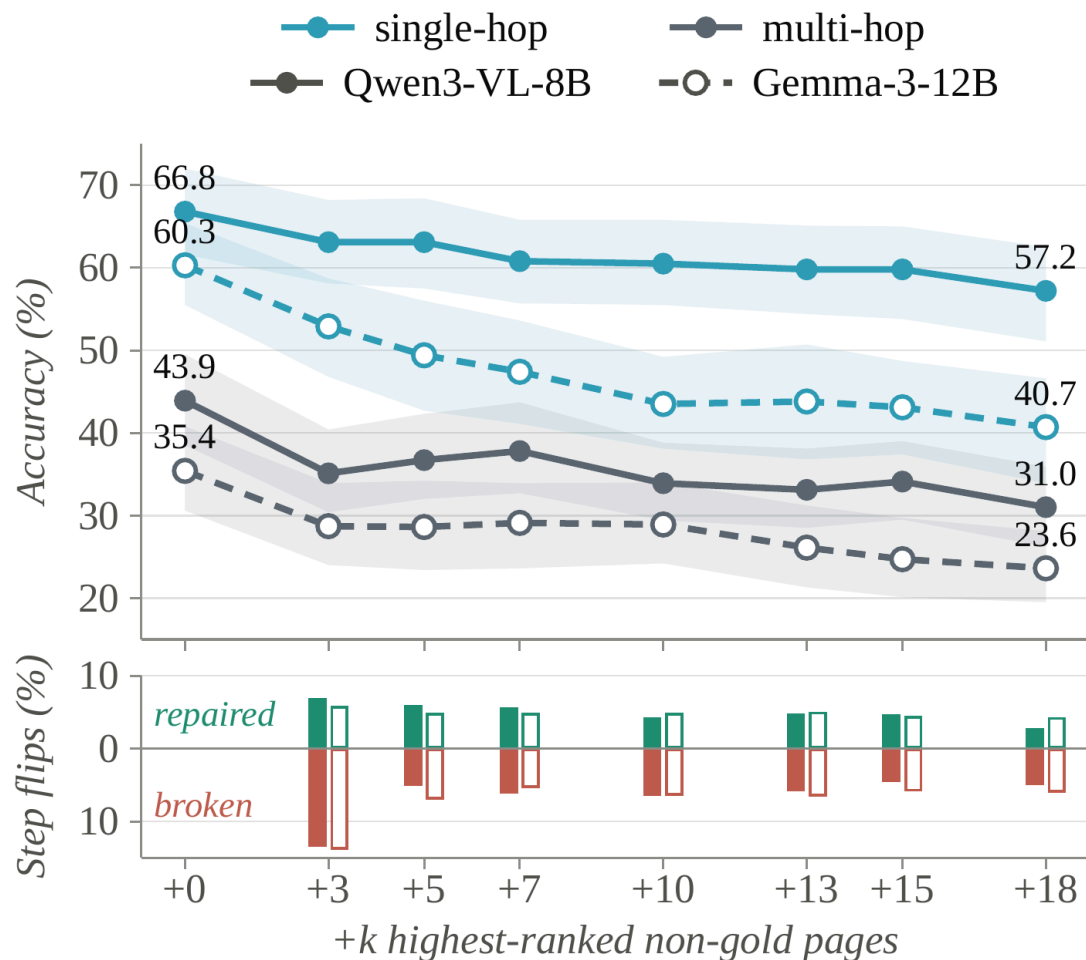


SELECTION

Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** (lowest-ranked) or 15.9% (highest-ranked).
- Correct answers break far more often than wrong ones are repaired.
- Retrieval rank does not predict which page is safe to lose.

Distractor Exposure



Encode
fixed

Retrieve
gold + k other pages

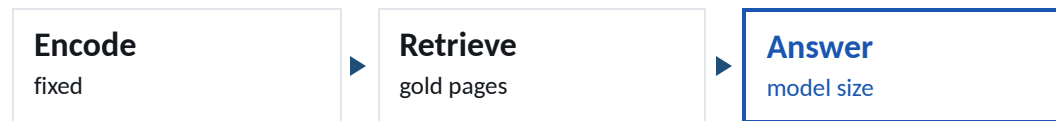
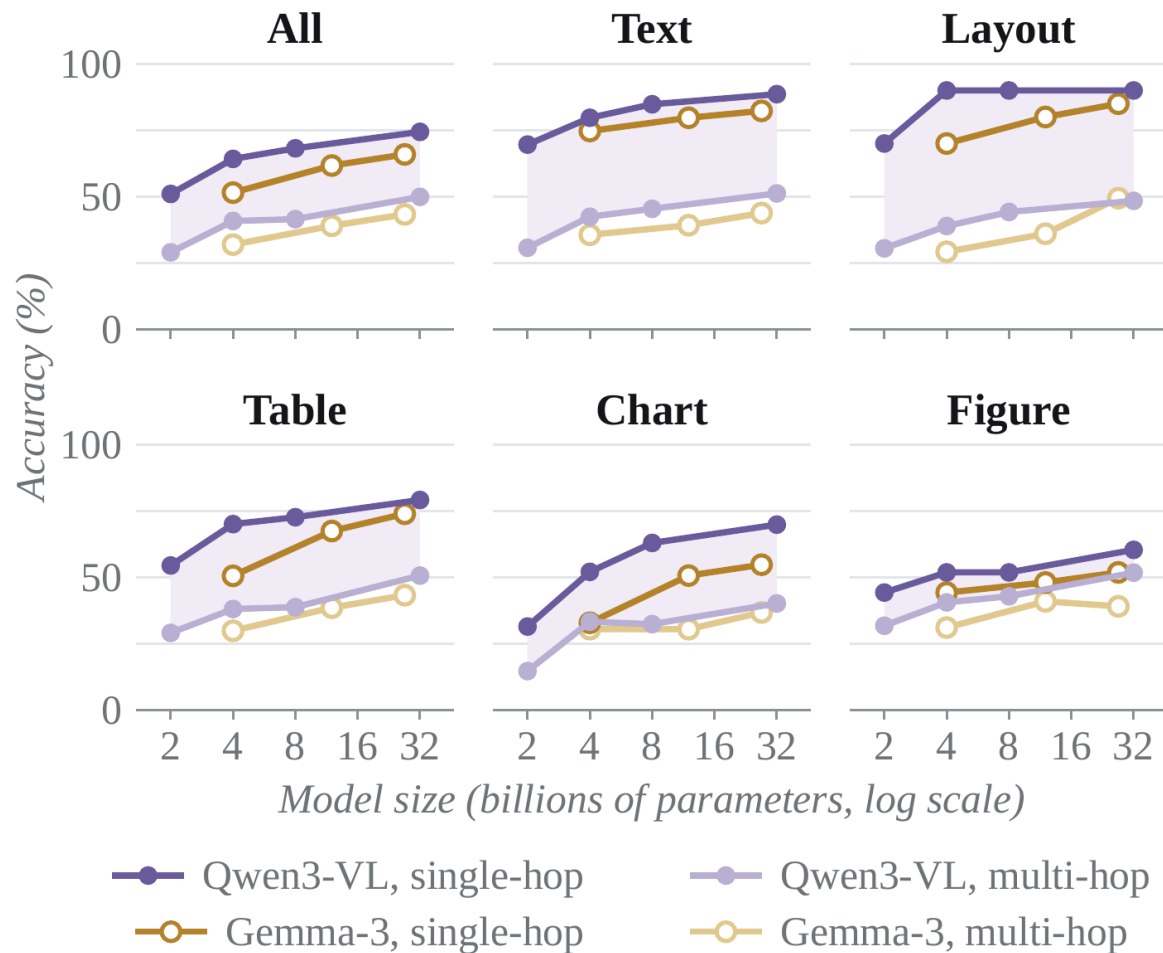
Answer
fixed

SELECTION

Multi-hop accuracy drops sharply with added context

- Qwen3-VL-8B: multi-hop falls from **43.9% to 35.1%** after three extra pages. Single-hop declines gradually, 66.8% to 57.2%.
- Gemma-3-12B shows the same trend: multi-hop 35.4% to 23.6%, single-hop 60.3% to 40.7%.
- Every gold page is still there. Extra context both breaks and repairs answers.

Evidence Integration

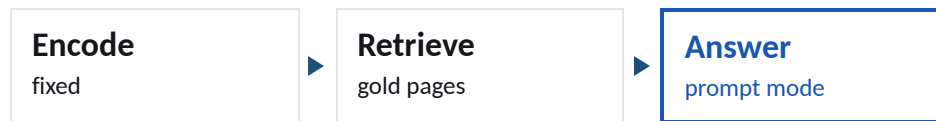
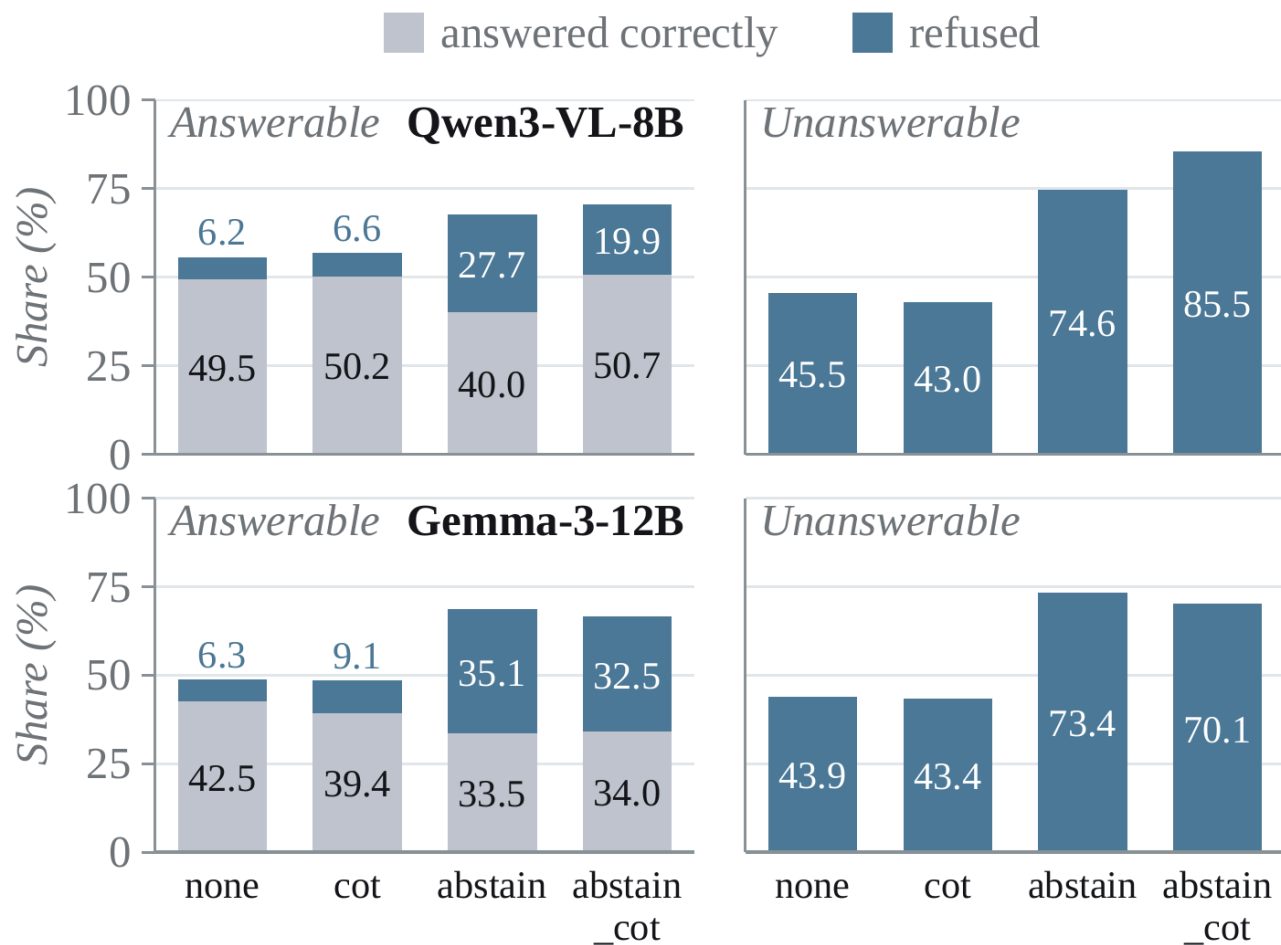


REASONING

Scaling helps, but the multi-hop gap persists

- Qwen3-VL from 2B to 32B: multi-hop rises from **29.0%** to **49.9%**, single-hop from 51.0% to 74.4%.
- The gap between them stays roughly the same.
- Gemma-3 shows the same separation.

Response Calibration



REASONING

Reasoning before abstaining limits false refusal

- Qwen3-VL-8B, abstain: unanswerable refusal **45.5% to 74.6%**, but false refusal **6.2% to 27.7%** and accuracy 49.5% to 40.0%.
- Adding CoT: false refusal 19.9%, accuracy 50.7%, unanswerable refusal 85.5%.
- Gemma-3-12B gains much less from CoT: false refusal 35.1% to 32.5%.

Deployment: Accuracy and Cost

Representation	Acc.	Text	Vis.	Tokens	Prefill	Decode
T	29.0	499	0	499	1.4	5.7
TL	35.7	874	0	874	1.8	6.6
TV	50.1	500	459	959	–	–
TLV	49.5	874	459	1333	26.3	6.7
V	42.6	20	459	479	29.0	7.1
Retriever	R@1	R@5	R@10	Idx. (s)	Qry. (ms)	GB
BM25	32.7	58.1	68.1	0.02	0.3	CPU
BGE-M3	32.8	61.3	72.4	0.01	47	1.2
Qwen3-Emb.	33.5	63.9	74.8	0.19	89	13.1
ColModernVBERT	47.1	74.0	82.7	0.91	41	5.1
ColQwen2.5	49.2	77.2	84.8	1.50	293	8.6
ColQwen3	53.7	81.0	87.9	0.57	110	9.4
Reasoner	T	TL	TLV	V	GB	Time (s)
Qwen3-VL-8B, bf16	31.9	38.4	52.4	45.5	17.5	30.7
8-bit	32.5	37.9	53.6	45.7	10.3~	43.5
4-bit	31.8	39.4	51.8	45.5	6.8~	35.4
Qwen3-VL-32B, 4-bit	37.0	42.1	59.1	52.3	20.0~	47.5
InternVL3-8B	19.3	25.3	32.6	24.1	15.9	5.3
MiniCPM-V-4.5	30.7	38.1	60.1	53.3	17.4	47.2
GLM-4.6V-Flash	25.9	32.9	42.7	30.9	20.6	19.8
Llama-3.2-11B-V	26.5	34.3	33.3	25.3	21.3	10.4
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4	8.0

PRACTICAL DISCUSSION

Every choice has a cost side

- **Representation:** embedded text plus images matches parser text plus images (50.1% against 49.5%) with fewer tokens. Images cost about 26 s of prefill against 1.8 s.
- **Retriever:** ColQwen3 reaches 81.0% recall at 5 against 58.1% for BM25, but needs a GPU and 110 ms per query against 0.3 ms.
- **Reasoner:** 4-bit keeps accuracy at a smaller footprint (51.8% in about 6.8 GB) but is not faster.

From Findings to Design

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION

Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

SELECTION

Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

REASONING

Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

Simple Method Cost

	Turns	Tokens			Latency (s)		Peak
		Input text	Input visual	Output	Prefill	Decode	VRAM (GiB)
reader	11.5	5,030	816	697	1.8	46.8	7.2
synthesizer	1.2	9,307	2,018	463	3.8	34.7	8.1
arbiter	1.2	6,806	1,734	464	2.9	33.3	7.9
Per question, mean	13.8	76,669	13,767	9,092	28.5	617.5	8.2
Per question, median	9.0	43,113	9,830	6,078	15.8	385.0	8.0

- 13.8 turns per question on average, 11.5 of them the reader.
- The cost is time, not memory: 10.8 minutes mean, 6.7 median, with peak memory at 8.2 GiB.